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PSYCHOLOGICAL SCALES ON SUICIDE, DEATH AND OTHER CONSTRUCTS DEVISED BY DAVID LESTER

Attribution of causes to suicide: David Lester & James Bean	2-5
Myths about suicide on the Suicide Opinion Questionnaire: David Lester, John McIntosh & James Rogers	6-8
A scale to measure the desire to be dead: David Lester	9-11
The Lester Attitude Toward Death Scale: David Lester	12-18
The Collett-Lester Fear of Death Scale: David Lester	19-22
Belief in a day of judgment and death anxiety: James A. Beshai & David Lester	23-26
Attitudes toward funerals. David Lester & Joshua Blustein	27-28
An inventory to measure helplessness, hopelessness, and haplessness: David Lester	29-32
An ontological insecurity scale: David Lester & Jeffrey Thinschmidt	33-35
The holistic theory of Andras Angyal: Measuring the system principle: David Lester & Benjamin Dench	36-39
Measuring Maslow's hierarchy of needs: David Lester	40-41
Attitudes toward life and aging: David Lester, Linda Monfredo & Holly Hummel	42-43
A short computer anxiety scale: David Lester, Bijou Yang & Simon James	44-47
Aversion to touching: David Lester & Alisha Rencher	48-49
The desire to lose one's identity: David Lester	50-53
The Shaw Blocks Test: David Lester	54-58
The Executive Personal Finance Scale: Marcello Spinella, Bijou Yang & David Lester	59-63

And, of course:

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ATTRIBUTION OF CAUSES TO SUICIDE

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In recent years, several scales have been developed to assess attitudes toward suicide. Domino, Gibson, Poling, & Westlake (1980) constructed a large inventory and identified clusters from a factor analysis of the items, tapping such areas as suicide as a sign of mental illness and suicide as a cry for help. In the present study, we sought instead to define several components of attitudes toward suicide and to measure those components.

The first set of components was related to the following types of causes to which people attribute suicide: intrapsychic problems ("People who commit suicide are usually mentally ill"); interpersonal conflicts ("Suicide is often triggered by arguments with a lover or spouse"); or societal pressures ("Those who are oppressed in a society are more likely to commit suicide"). In addition, two sets of items were constructed to measure the individual's personal approval ("I believe that suicide can be a rational act") or disapproval ("Only cowards kill themselves") of suicide and the attitudes of the individual's peer and cultural groups ("Some of my friends think that suicide is an acceptable choice"). (The scales may be obtained by writing to the senior author.)

The items were combined and given anonymously in a Likert-type format to 27 men and 45 women aged 18 to 23 years (mean age, 20.1, standard deviation, 1.5) who were enrolled in college undergraduate courses in the United States. They were also given a test of general irrational thinking based on Ellis's rational-emotive therapy (Lee, Hallberg, & Haase, 1979) and a test of belief in an external locus of control (Rotter, 1966).

Beliefs in the three causes of suicide were all weakly or moderately associated. For intrapsychic problems and interpersonal conflicts (Pearson's $r=.20$, $p=.05$) for intrapsychic problems and societal pressures ($r=.25$, $p=.02$) and for interpersonal conflicts and societal pressures ($r=.32$, $p=.003$). Thus, the critical dimension here seems to be the belief that suicide has definite causes regardless of the type of cause. Having a negative personal view of suicide was associated only with the strength of the belief that suicide is caused by intrapsychic problems ($r=.36$, $p=.001$).

Negative attitudes of the peer group and culture toward suicide were not associated with beliefs about the causes of suicide but were associated with a negative personal attitude toward suicide, ($r=-.38$, $p=.001$) suggesting that the peer group and culture may shape the individual's evaluation of suicide. It would be interesting in future research to explore how the beliefs of close family members and significant others affect one's own beliefs about suicide.

The two measures of personality were generally unrelated to beliefs in the causation of suicide or attitudes toward suicide. Belief in intrapsychic causes was only weakly associated with irrational thinking. These findings may reflect the particular traits used in this study, but they may also reflect the possibility that attitudes toward suicide are rooted in factors other than personality traits.

These five new scales will, we hope, stimulate research into attitudes toward suicide and illustrate the usefulness of first defining dimensions of attitudes and then devising scales to measure these dimensions.

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Belief in intrapsychic causes

1. People who commit suicide are usually mentally ill.
2. People often commit suicide as a way of punishing themselves
3. Most people who kill themselves are depressed.
4. People who survive an attempt to kill themselves should be required to have psychotherapy so that they can understand their inner motivations.
5. People who kill themselves have lost their belief in God.
6. People who kill themselves are thinking irrationally.

Belief in interpersonal causes

1. Suicide is often triggered by arguments with a lover or spouse.
2. Suicide can be caused by work stress such as being fired from your job.
3. Experiencing a broken home as a child is likely to increase your risk of suicide.
4. The way a person was raised as a child is probably unrelated to whether he or she will later kill him/herself.
5. When someone kills him/herself, you usually find that the family and friends are to blame.
6. A suicide attempt is usually an attempt to get sympathy from others.

Belief in societal causes

1. Those who are oppressed in a society are more likely to commit suicide.
2. People who live under dictatorships are more likely to kill themselves.
3. People who face starvation, malnutrition and the accompanying disease are likely to kill themselves.
4. If a culture were to allow the open expression of feelings such as anger or shame, then people would be less likely to kill themselves.
5. There are features of modern industrialized societies which make suicide more common.
6. Suicide among youth is more likely these days because of the tremendous problems facing the world such as pollution and the threat of nuclear war.

Personal attitudes

1. People who commit suicide are basically weak people.
- *2. I believe that suicide can be a rational act.
3. Only cowards kill themselves.
4. Most suicides are psychiatrically disturbed.
- *5. Suicide is acceptable for people under great personal stress.
6. Suicide is a sinful act
- *7. Doctors should be able to give a patient a lethal dose if that patient wants and asks to die.

- *8. It is acceptable for people dying of an incurable disease to take their own life.
- *9. There may be situations where suicide is the only reasonable choice.
- 10. People should be prevented from committing suicide since they are not acting rationally at the time.

Normative beliefs

- 1. The religion I was raised in views suicide as a sin.
 - 2. My mother has a negative view of suicide.
 - *3. I have met teachers who think that suicide can be a rational action.
 - *4. Some of my friends think that suicide is an acceptable choice.
 - 5. My father has a negative view of suicide.
- * positively keyed items

MYTHS ABOUT SUICIDE ON THE SUICIDE OPINION QUESTIONNAIRE: AN ATTEMPT TO DERIVE A SCALE

David Lester, John McIntosh & James Rogers

The identification of "myths about suicide" has been of interest to suicidologists as a test of accurate knowledge about suicide in evaluating suicide education programs (e.g., Kalafat & Elias, 1994). Domino, Moore, Westlake, and Gibson (1982) published the Suicide Opinion Questionnaire, a 100-item scale to measure attitudes toward suicide, and Domino (1990) identified 20 of those items which textbooks commonly report to be popular misconceptions about suicide. Perusal of these 20 items by three judges (the present authors) suggested that not all of these items were misconceptions about suicide. The present study was designed to identify which of the 100 original items might be clearly true or false rather than simply opinions about suicide.

The present authors made judgments as to whether each of the 100 items of the Suicide Opinion Questionnaire was true or false based on available research or clinical evidence. The three judges agreed on seven items:

1. Item 3 false: The suicide rate is higher for blacks than for whites.
2. Item 12 true: In the U.S. suicide by shooting oneself is the most common method.
3. Item 22 true: Suicide rates vary greatly from country to country.
4. Item 30 false: Over the past ten years the suicide rate in this country has increased greatly.
5. Item 64 true: A person whose parent has committed suicide is a greater risk for suicide.
6. Item 91 false: The large majority of suicide attempts result in death.
7. Item 96 true: Most people who attempt suicide fail in their attempts.

Only one of these seven items (Item 64) is among the 20 items in Domino's list of misconceptions reported by textbooks. A further eight items were judged similarly by two of the three judges, with the third judge indicating that he could not make a judgment. (The remaining items had only one judge making a definite judgment or had conflicting answers from the judges.)

1. Item 2 true: Almost everyone has at one time or another thought about suicide.
2. Item 6 false: Most suicides are triggered by arguments with a spouse.
3. Item 20 true: Some people commit suicide as an act of self-punishment.
4. Item 44 true: The possibility of committing suicide is greater for older people (those 60 and over) than for younger people (20-30).
5. Item 48 false: Once a person is suicidal, he is suicidal forever.

6. Item 52 false: Improvement following a suicidal crisis indicates that the risk is over.
7. Item 86 false: Suicide occurs only in civilized societies.
8. Item 98 true: Individuals who are depressed are more likely to commit suicide.

Only three of these eight items (Items 48, 52, and 98) are among the 20 items listed by Domino as misconceptions reported by textbooks. Thus, 16 of the items from the 20 textbooks did not elicit agreement from the present three judges as to their truth or falsity.

Past researchers have occasionally used "myths about suicide" scales, and the seven and fifteen (7 + 8) items identified here can be considered to form two such scales. To explore the conventional reliability of these scales, the Suicide Opinion Questionnaire was administered to 40 male and 156 female undergraduate students enrolled in social science courses at a state college (Mage = 23.0, SD = 5.6). Cronbach alpha for the 7-item scale was 0.07 and for the 15-item scale 0.16. Thus these two myths about suicide scales have poor reliability.

On only two items were the majority of respondents incorrect. Respondents thought that suicide rates had increased greatly in recent years (Item 30) and that those in their 20s had higher suicide rates than those over the age of 60 (Item 44).

Incidentally, scores on the 7-item myth scale were not significantly associated with sex or scores on Templer's Death Anxiety Scale (Templer, 1970), the four subscales of the Collett-Lester Fear of Death Scale (Lester & Abdel-Khalek, 2003), and Reynolds' Suicide Ideation Questionnaire (1987); Pearson r 's = 0.05, -0.02, 0.04, 0.14, 0.07, 0.11, and 0.06, respectively. Scores on the myth scale were weakly associated with age ($r = -0.16$, two-tailed $p < .01$).

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A SCALE TO MEASURE THE DESIRE TO BE DEAD

David Lester

There are many scales to assess current suicidality, such as the Beck Suicide Ideation Scale (Beck, et al., 1979), but Marušić, et al. (2012) noted that intervention would be better if it could be made earlier in the development of the suicidal process. Marušić and Farmer (2001) proposed that general negative thoughts (e.g., ideation about death) and passive suicide ideation (e.g., wishing to be dead) are the beginning phases of the suicidal process, and these stages are not assessed by most of the currently available suicide assessment tools. Marušić and his colleagues argued that suicide prevention would benefit from tools that cover these stages and they would enable the suicidal process to be followed more closely. A person with suicidal ideation who has a long history of wishing to be dead might be at greater risk of committing suicide than a person who does not have a history of wishing to be dead.

A similar point was made by Cavan (1928) in her classic book on suicide. She noted that the desire to never have been born is the most vague of all death wishes and does not necessarily accompany a desire to commit suicide. Cavan reported the case of an American female college student who started to have this wish at age 15, accompanied by feelings of despair and hopelessness. She was usually alone when the wish came to her, and she would say to herself, "Why was I born if I am always to be making clumsy mistakes?" (p. 180). Cavan also reported cases of individuals who day-dream of being dead, and she noted that these individuals clearly have favorable attitudes toward suicide and they may be thought of as potential suicides. She entitled her chapter on these cases *Incipient Suicides*.

Ayalon and Shiovitz-Ezra (2011) noted that as many as 7% of primary care patients with no depression report death ideation which is not synonymous with suicidal ideation. In a European sample of people in the second half of life, many reported having had passive death wishes in the prior month ("In the past month, have you felt that you would rather be dead"). In those over the age of 75, 15.6% agreed with this statement, 7.3% of those aged 66-75, and 4.6% of those aged 50-65, and having such a wish was associated with loneliness in the two younger groups.

Baca-Garcia, et al. (2011) noted that the desire for death is not generally considered to be a harbinger of more severe suicidal behavior and is not routinely included in suicide research or in risk assessment. They found that those who desired death but did not have suicidal ideation when assessed had as high a rate of lifetime attempted suicide as those with suicidal ideation and no desire for death. Those with both had the highest lifetime rate of attempted suicide. However, it appears that Baca-Garcia et al. used a single question to measure the desire for death.

Some scales to measure suicidal ideation include questions relevant to both the desire to be dead and suicidal ideation, which prevents separate assessment of each construct. For example, the Suicide Ideation Questionnaire (SIQ) (Reynolds, 1987) has 30 items, including some that ask about suicide ideation ((I thought about killing myself) and some about the desire to be dead (I wished I were dead), as well as items on other issues (e.g., I thought about having a bad accident). Although the SIQ has been used in many research studies, a total score is always used rather than teasing out scores for the desire to be dead and suicidal ideation separately.

The present article reports an attempt to derive a scale to measure the wish to be dead, separate from the wish to commit suicide, so that a more precise measure may be obtained than from a single question.

Method

Eleven items were written to assess the wish to be dead, and administered to 96 female and 16 male undergraduates enrolled in psychology courses (mean age = 22.4 years, SD = 4.4) using a true/false answer format.

Results

Item-total correlations were examined, and one item had the weakest correlations (all statistically non-significant) with the other items (I sometimes dream about being dead) and was discarded, leaving 10 items in the scale (see Table 1). The Cronbach alpha for the 10-item scale was 0.83. (Adding the 11th item reduced the alpha to 0.82.)

A principal components analysis with a varimax rotation to extract one factor indicated that all the items loaded on this single factor with factor loadings ranging from 0.36 to 0.83, with a median of 0.65 (see Table 1). Thus, the scale items seem to be reasonably homogeneous.

Table 1. The Items of the Wish to be Dead Scale

	Loadings
There have been times when I wished that I were dead.	0.67
I sometimes think that death would solve my problems.	0.71
Sometimes I wish I could go to sleep for several years.	0.44
I occasionally day-dream about being dead.	0.36
I sometimes think that there is little point in living.	0.68
I have occasionally fantasized about my funeral.	0.49
I have on occasions lost my desire to live.	0.83
I sometimes wish that I had never been born.	0.57
I sometimes think that there is no purpose to life.	0.63
It occasionally crosses my mind that life is not worth living.	0.82
% of variance	40.7%

Forty students reported a history of suicidal ideation, and the point-biserial correlation between a history of suicidal ideation and the total wish to be dead scale score was 0.57 (two-tailed $p < .001$). Eight students reported a previous suicide attempt and the point-biserial correlation between a history of attempted suicide and the wish to be dead was 0.34 ($p < .001$). The students were also given a 9-item death obsession scale (Abdel-Khalek & Lester, 2003) and the Pearson correlation between the total wish to be dead scale score and the death obsession score was 0.37 ($p < .001$), indicating that, although the two variables are related, the correlation is only modest. Thus, the wish to be dead scale appears to be measuring a variable distinct from an obsession with death.

Twenty-five of these same students were given the scale measuring the wish to be dead again 2 weeks later. The 2-week test-retest correlation was 0.87 ($p < .001$). Thus, the scale appears to have good test-retest reliability.

Discussion

The 10-item scale to measure the wish to be dead, without mentioning suicide at all, appears to have moderate reliability. The test-retest reliability and Cronbach alphas were satisfactory for a brief scale. Furthermore, although scores on the scale correlate modestly with death obsession and with a history of suicidal ideation and attempting suicide, the correlations were modest, indicating that the scale may be measuring a construct different from either of those.

Marušić, et al. (2012) proposed that the wish to be dead assessed a state of mind on the pathway to suicide, but at a point earlier on in the suicidal process, and that knowledge of this state of mind might provide an early warning sign for therapists treating potentially suicidal clients. The scale presented here measures the wish to be dead separately from suicidal ideation and, therefore, is a useful addition to the scales for assessing steps on the pathway to suicide. The results of the present study also suggest that scales to assess suicidal ideation should focus on suicidal ideation per se and eliminate items relevant only to the desire to be dead. Some scales to assess suicidality currently in use have items pertinent to both constructs.

The present exploratory study had several limitations. First, the sample was from anon-clinical population (college students), and it is important to explore the use of the scale in a clinical population such as psychiatric outpatients. Second, scores on the scale were studied for their association with past suicidality, and the association of the scores with current suicidality would be of interest. Third, a prospective study would help determine whether the desire to be dead is a step on the path leading to suicidal ideation and, eventually, overt suicidal behavior (non-lethal and lethal). However, the aim of the study was to develop a scale to measure the construct proposed by Maruko and his colleagues so that this construct can be studied for its use in the clinical evaluation of potentially suicidal individuals, and the present report achieved this goal. Further research is needed to see if the scale proposed here has value, both in non-clinical and

clinical populations, but it is hoped that the publication of this scale will stimulate further research on this important topic.

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The Items

For each of the following items, please answer whether it is generally true for you using the following scale. Please answer every item.

SA strongly agree

A agree

MA mildly agree

MD mildly disagree D disagree

SD strongly disagree

1. There have been times when I wished that I were dead.
2. I sometimes think that death would solve my problems.
3. Sometimes I wish I could go to sleep for several years.
4. I occasionally day-dream about being dead.
5. I sometimes think that there is little point in living.
6. I have occasionally fantasized about my funeral.
7. I have on occasions lost my desire to live.
8. I sometimes wish that I had never been born.
9. I sometimes think that is no purpose to life.
10. It occasionally crosses my mind that life is not worth living.

THE LESTER ATTITUDE TOWARD DEATH SCALE

David Lester

Lester devised an attitude toward death scale that has been used in many studies. The present presents the scale and reviews some research evidence for its reliability and validity.

The Scale

The scale was devised as an equal-interval scale in which statements about an attitude, in this case an attitude toward death, are scaled using judges' ratings to indicate the degree of favorability toward death. Then, the median value of the items a respondent agreed to becomes the score for that respondent. The final scale consists of twenty-one items (the scale is shown in Appendix A,

The Lester Attitude Toward Death Scale is unique among fear of death scales in that two parallel forms were devised. The parallel forms are shown in Appendix B.

It should be noted that the scale permits both a measure of the average attitude toward death and a measure of the range of attitudes agreed with, which Lester suggested could serve as a measure of the inconsistency of the respondent's attitude toward death. Again, no other fear of death scale permits such a measure.

Method of Construction

Ninety-eight statements on the subject of death were collected, numbered, and typed in book form. The statements were given to 22 students in an Experimental Psychology course who were asked to judge each statement on an 11-point scale for the degree to which each statement reflected a favorable attitude toward death. The resulting data were used to construct two equal-interval scales using the procedure described by Edwards (1957).

Reliability

Parallel Forms

The Spearman rank correlation between the two parallel forms using a sample of eighteen undergraduates was 0.65.

Test-Retest

The Spearman rank correlation for 14 students tested six weeks apart was 0.58. For 80 tested six weeks apart, the Spearman correlation was 0.70 for the attitude toward death score and 0.69 for the inconsistency score.

Attitudes toward Death and Inconsistency

Lester reported that, of seventeen correlations between the attitude toward death score and the inconsistency score, eleven were positive and six negative. Fifteen were not significant, while two were significant and positive. Thus, any association, if one exists, is quite weak.

Social Desirability

Scores on the Lester scale do not appear to be associated with scores on a scale of social desirability.

Other Research¹

For 40 students, the arithmetic mean and the median score of the statements agreed with were highly correlated (Pearson $r = 0.93$). In addition, those who omitted an item on the Lester scale did not differ in their scores on a fear of death scale from those who completed every item on the Lester scale. Lester found that the average score on the Lester scale was not associated with the number of items agreed with.

Validity

Concurrent Validity

In the initial construction of the scale, scores on the scale were found to be associated with students' ratings of their own fear of death on a graphic rating scale (Spearman $\rho = 0.44$, $n = 22$). Scores were also associated with scores on, and with responses to a question on belief in life after death.

Four studies have correlated scores on the scale with scores on the four sub-scales of the Collett-Lester Fear of Death Scale. The median correlation out of the twelve correlation coefficients was 0.57 for fear of death of self,

¹ See the References for citations to my research on this scale.

0.05 for fear of death of others, 0.29 for fear of dying of self and 0.17 for fear of dying of others. These averages suggest that the Lester scale is primarily a measure of the fear of one's own death.

Factor Analysis of Fear of Death Scales

Two studies have factor-analyzed scores from a set of fear of death scales. Both reported that the Lester scale score is associated with a general fear of death, in particular, of one's own death.

Correlates of the Attitude toward Death

Scores on the Lester scale were associated with having less of a purpose/meaning in life, a history of suicidal thoughts/threats, a poor memory for dreams, and one's attitude toward life, alienation, and animistic beliefs.

Males were reported to have higher scores than females. More experienced nursing students and faculty had less fear of death and less inconsistency, though specialty was not related to scores. Lester found that staff at a suicide prevention center had fewer inconsistent attitudes (but not different average attitudes) than staff at other types of mental health clinics. He also found that religiosity was associated with more favorable attitudes toward death and more consistent attitudes.

Discussion

The results obtained by researchers in their studies of death attitudes differ greatly depending upon the type of approach they choose. The difficulty in devising and using disguised measures and subliminal measures of death attitudes has resulted in almost all of the current measures in use being of the face-valid, self-report type.

Given this focus, it is important to devise, not simply alternative measures of the same construct, but measures of different constructs. The devising of fear of death scales with different subscales and the factor analysis of existing scales to identify clusters of related items has been useful in this regard. The present Lester scale for attitudes toward death is unique in offering a measure of the inconsistency of death attitudes in a respondent would be useful for researchers.

The Lester scale appears to measure the fear of personal death, and in this respect, it is similar to several other published scales. However, the use of an equal-interval scale with assigned weights to individual items is unique.

Most other fear of death scales use a simple agree/disagree format (often with degrees of disagreement/agreement included). The Lester scale offers an alternative measurement technique for assessing the fear of personal death.

These concerns with measurement are not irrelevant to theory and research findings, for the particular measure used may greatly affect the outcome of research and, thereby, support for a particular theory. As a result, many investigators now routinely use several fear of death scales in order to show that their results are not dependent on the use of one particular scale.

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APPENDIX A: The Lester Attitude toward Death Scale
(The numbers are the item scores and should not be put on the questionnaire)

Please circle A or D for each of the following statements. If you agree with the statement, circle A. If you disagree, circle D. Try to circle a reply for each statement. Consider the death in each statement to mean your death at the present time.

- 1 A D What we call death is only the birth of the soul into a new and delightful life (1.29)
- 2 A D One should not grieve over the dead, because they are eternally happy in heaven (2.00)
- 3 A D Death comes to comfort us (2.81)
- 4 A D Death will be one of the most interesting experiences of my life (3.25)
- 5 A D A peaceful death is a fitting end to a successful life (3.75)
- 6 A D I don't want to die right now, but I'm glad that I will die someday (4.12)
- 7 A D Death is better than a painful life (4.59)
- 8 A D I would be willing to die to save my best friend (5.00)
- 9 A D Death makes all men equal (5.11)*
- 10 A D Death is a great mystery (5.58)
- 11 A D Death is neither good nor bad since there's no consciousness in it (6.18)
- 12 A D You can't take it with you when you die (6.85)
- 13 A D I would feel better about death if I knew what it was going to be like (7.17)
- 14 A D It is a pity when a talented man dies, even if he has stopped creating (7.42)
- 15 A D Death is an unwanted sleep (7.92)
- 16 A D Death is to be feared for it brings grief (8.15)
- 17 A D I am afraid to die because there may be a future punishment (9.42)
- 18 A D Nothing can be so bad that a sane man would commit suicide (9.80)
- 19 A D Death is the last and worst insult to man (10.19)
- 20 A D I would avoid death at all costs (10.42)
- 21 A D Death is the worst thing that could possibly happen to me (10.76)

Note: The scale was devised in 1966. For use today, the sexist language should be eliminated

APPENDIX B: The Alternative Forms
(The numbers are the item score and should not be put on the questionnaire)

Form A

1. Death is utopia (1.23)
2. One should not grieve over the dead because they are eternally happy in heaven (2.00)*
3. Death could be a lot of fun (2.64)*
4. Death comes to comfort us (2.81)
5. Death will be one of the most interesting experiences of my life (3.25)
6. Death is the spice of the adventurer (3.91)
7. Death is better than a painful life (4.59)
8. I would rather be put to death than spend my life in jail (4.85)
9. Death makes all men equal (5.11)
10. Death makes rich of the poor and poor of the rich (5.88)
11. Death is neither good nor bad, since there's no consciousness in it (6.18)
12. Death is the final battle of life (6.81)
13. I would feel better about death if I knew what it was going to be like (7.17)
14. It is a pity when a talented man dies, even if he has stopped creating (7.42)
15. Death is an unwanted sleep (7.92)
16. Death is the thief that robs us of life (8.19)
17. Abortion is wrong, for even a child in its mother's womb has a soul (9.35)
18. Death brings all one's plans, all one's hopes to an end (9.59)
19. Death is the last and worst insult to man (10.19)
20. If wounds are painful, death and the dissolution of the body must be an agony (10.64)

Form B

1. What we call death is only a birth of the soul into a new and delightful life (1.29)
2. One should not grieve over the dead because they are eternally happy in heaven (2.00)*
3. Death could be a lot of fun (2.64)*
4. Death brings security to us all (3.00)
5. A peaceful death is a fitting end to a successful life (3.75)
6. I don't want to die right now, but I'm glad that I will die some day (4.12)
7. Death is welcomed by the hospitable soul (4.61)
8. I would be willing to die to save my best friend (5.00)
9. Death is a great mystery (5.38)

10. Death is with us from the time that we are born (6.01)
11. Since nothing can be done about death, there is no reason to think about it (6.35)
12. You can't take it with you when you die (6.85)
13. I find it difficult to believe that I will be dead some day (7.23)
14. Death leaves life unfinished (7.72)
15. Death is to be feared for it is beyond our control (8.00)
16. Death is to be feared for it brings grief (8.15)
17. I am afraid to die because there may be a future punishment (9.42)
18. Nothing is worth dying for (10.00)
19. I would avoid death at all costs (10.42)
20. Death is the worst thing that could possibly happen to me (10.76)

*These items are repeated in both forms since few other items received scale values in this range.

THE COLLET-LESTER FEAR OF DEATH SCALE

David Lester

There have been several versions and publications about the Collet-Lester Fear of Death Scale (CLFDS). The CLFDS is unique in measuring four separate fears:

- Death of self
- Dying of self
- Death of others
- Dying of others

Unlike some fear of death scales, it does not have items relating to funerals or cemeteries. Lester and Blustein devised a scale to measure these aspects of death.

The following is a list of Lester's articles on the CLFDS, followed by an Appendix with the final version of the scale. Lester typically gives the scale as presented below rather than mixing the four sets of items.

- Abdel-Khalek, A., & Lester, D. (2004). The factorial structure of the Arabic version of the revised Collett-Lester Fear of Death Scale. *Death Studies*, 28, 787-793.
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Your age: ____ years

Your sex:

____ male

____ female

How disturbed or made anxious are you by the following aspects of death and dying? Read each item and answer it quickly. Don't spend too much time thinking about your response. We want your first impression of how you think right now. Circle the number that best represents your feeling of upset/disturbed/anxious.

Your own Death		not at all	somewhat	very much		
1. The total isolation of death		1	2	3	4	5
2. The shortness of life		1	2	3	4	5
3. Missing out on so much after you die		1	2	3	4	5
4. Dying young		1	2	3	4	5
5. How it will feel to be dead		1	2	3	4	5
6. Never thinking or experiencing		1	2	3	4	5
disintegration of your body after you die	1	2	3	4	5	57. The

Your Own Dying

1. The physical degeneration involved		1	2	3	4	5
2. The pain involved in dying		1	2	3	4	5
3. The intellectual degeneration of old age		1	2	3	4	5
4. That your abilities will be limited as you lay dying		1	2	3	4	5
5. The uncertainty as to how bravely you will face the process of dying.		1	2	3	4	5
6. Your lack of control over the process of dying		1	2	3	4	5
7. The possibility of dying in a hospital away from friends and family		1	2	3	4	5

The Death of Others

1. Losing someone close to you		1	2	3	4	5
2. Having to see the person's dead body		1	2	3	4	5
3. Never being able to communicate with the person again		1	2	3	4	5
4. Regret over not being nicer to the person when he or she was alive		1	2	3	4	5
5. Growing old alone without the person		1	2	3	4	5
6. Feeling guilty that you are relieved that the person is dead		1	2	3	4	5

7. Feeling lonely without the person	1	2	3	4	5
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The Dying of Others

1. Having to be with someone who is dying	1	2	3	4	5
---	---	---	---	---	---

2. Having the person want to talk about death with you	1	2	3	4	5
--	---	---	---	---	---

3. Watching the person suffer from pain	1	2	3	4	5
---	---	---	---	---	---

4. Seeing the physical degeneration of the person's body	1	2	3	4	5
--	---	---	---	---	---

5. Not knowing what to do about your grief at losing the person when you are with him/her	1	2	3	4	5
---	---	---	---	---	---

6. Watching the deterioration of the person's mental abilities	1	2	3	4	5
--	---	---	---	---	---

7. Being reminded that you are going to go through the experience also one day	1	2	3	4	5
--	---	---	---	---	---

BELIEF IN A DAY OF JUDGMENT AND DEATH ANXIETY

James A. Beshai & David Lester

In studies of the relationship between religiosity and death anxiety, one important component of religious beliefs has been omitted from consideration, namely one's belief in a day of judgment after death. The present study was designed to investigate whether there are correlations between belief in a day of judgment, religiosity, and the fear of death.

Previous research on religiosity and death anxiety has found no association (e.g., Abdel-Khalek & Lester, 2009) or only a weak association (e.g., Ens & Bond, 2007). Lonetto and Templer (1986) observed that the absence of a significant relationship between the Death Anxiety Scale and religious variables is inconsistent with the impressions of clinicians who claim that more religious persons, in the traditional sense, have lower death anxiety scores. It may be that religious individuals who believe in a day of judgment may differ in death anxiety from those who do not believe in such a day and, therefore, belief in a day of judgment may act as a moderator in the relationship between religiosity and death anxiety.

Belief in a day of judgment emerged in ancient Egypt about 2,400 B.C., where it found its most elaborate expression (Brandon, 1967). Preoccupation with life and death and a day of judgment is found in all religious texts but most notably in Christianity and Islam. The concept of life as a gift exchange is expressed in the scriptures in various forms, such as the Ten Commandments, the Sermon on the Mount, and several chapters on judgment in the Qur'an. Since belief in a day of judgment is present in many religions and held by many people, the present research was designed to devise a measure of the belief in a day of judgment (to our knowledge for the first time) and to explore whether belief in a day of judgment and religiosity are associated with the fear of death.

Method

Questionnaires were given to 31 male and 131 female undergraduate students in psychology courses at a rural state college (M age = 20.8 yr., SD = 4.1) with institutional review board approval. All participants were given the revised version of the Collett-Lester Fear of Death Scale (Lester & Abdel-Khalek, 2003). This has four subscales, each with seven items, answered on a 5-point Likert-type scale with anchors of 5: Very and 1: Not at all, for how distressed, disturbed, or anxious they are by aspects of death of self (e.g., the total isolation of death), dying of self (e.g., the pain involved in dying), death of others (e.g., losing someone close to you), and dying of others (e.g., watching the person suffer from pain). Mean scores were: fear of death of self, 22.0 (SD = 6.8), fear of dying of

self, 24.3 ($SD=6.2$), fear of death of others, 26.9 ($SD=5.1$), and fear of dying of others, 26.2 ($SD = 5.8$).

Seventy-seven of these 162 students were given Francis's Religiosity Scale (Francis, 1992), which has 24 items answered on a 6-point Likert-type scale, with anchors 1: Strongly agree and 6: Strongly disagree. A typical item is, "I think that going to church is a waste of time." The mean score was 90.6 ($SD = 33.9$).

Eighty-five of the 162 students were also given a scale developed to measure belief in a day of judgment. For this scale, 20 items were written to tap this belief (see the Appendix) and presented in a Likert-type format with anchors 1: Strongly disagree and 5: Strongly agree. The mean score for the 20-item scale was 64.6 ($SD=16.0$).

Results

The 20-item Scale

Cronbach's alpha reliability coefficient for the 20-item Belief in a Day of Judgment scale was 0.92. A principal components extraction (PCA) with Varimax rotation identified four components (Table 1). It can be seen that 12 of the items loaded strongly ($>.65$) on the first component, and it may be preferable to reduce the Belief in a Day of Judgment Scale to these 12 items. Certainly the cross-loadings should be examined.

Scores on the Belief in a Day of Judgment scale were not related to the fears of death and dying (Pearson r s ranged from -0.08 to 0.06), nor were scores on the Religiosity Scale (r s ranged from -0.01 to 0.17). To see if scores on the Religiosity Scale and the Belief in a Day of Judgment scale were associated, the 20-item Belief in a Day of Judgment Scale and Francis's Religiosity Scale were administered to a new sample of 10 male and 41 female undergraduates in psychology courses at the same college (M age=22.6 yr., $SD=4.4$). The Pearson correlation between scores on the two scales was 0.79 (two-tailed $p < .001$).

The 12-item Scale

The 12-item Day of Judgment Scale (Items 1–5 and 7–13 in the Appendix, p.5) had a mean score of 3.8 ($SD=13.0$) and a Cronbach alpha of 0.96. The Pearson correlations with the scores on the Collett-Lester Fear of Death Scale were: death of self 0.01, dying of self -0.01, death of others -0.12, and dying of others -0.15, all non-significant.

Table 1: Principle Components analysis of the 20-item scale

Item	Factor			
	1	2	3	4
1	.86	.06	-.08	.03

2	.79	.22	-.04	-.12
3	.78	.08	-.04	.01
4	.91	.01	.08	.01
5	.75	.23	-.04	-.03
6	.09	-.06	-.02	.86
7	.67	.33	.01	-.22
8	.86	.17	-.01	.01
9	.84	-.01	.17	.02
10	.75	.17	-.08	.07
11	.91	.09	.03	-.02
12	.87	.06	.14	.12
13	.78	.32	.09	.18
14	.54	.68	-.18	.00
15	.43	.64	.14	.05
16	-.02	.44	.53	.36
17	.07	.64	.50	.06
18	.04	.74	.07	-.13
19	-.08	.07	.89	.07
20	.17	.05	.72	-.44

% of variance 46.0 12.2 6.6 6.0

Eigenvalues 9.20 2.43 1.31 1.20

Note: Numbers in bold represent loadings >.40.

Discussion

The Belief in a Day of Judgment Scale appears to have good reliability, but reducing the scale to the 12 items with high loadings on the principal components analysis may be more meaningful in future research, since reviewers thought that some of the items lacked face validity and many had cross-loadings. Although moderately correlated with scores on a traditional religiosity scale, the Belief in a Day of Judgment Scale appears to tap a different religious belief. Interestingly, belief in a day of judgment appeared not to be associated with fears of death and dying, nor was religiosity, consistent with previous studies. The study has several limitations, including the preponderance of female subjects and the omission of a question assessing religious affiliation. However, it is hoped that the publication of this scale will facilitate and encourage future research on this particular belief.

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APPENDIX

Please respond to each item by a score from 1: Strongly disagree to 5: Strongly agree. The scale is an attempt to determine the degree of one's belief in a judgment of one's behavior as being accountable in some form of life after death. Feel free to agree or disagree with the concept of the Day of Judgment in a life hereafter.

- *1 God is the savior and judge of mankind.
- *2 If I am pure of heart and hands, I shall live in the fear of a Just God.
- *3 The whole cosmos speaks of the glory of God.
- *4 On the Day of Judgment, God shall judge the living and the dead.
- *5 Only God knows the hour and place where one will die.
- 6 Human beings are mortal, that is, they die and become dust.
- *7 God does not allow good deeds to go unrewarded.
- *8 I believe in a Day of Judgment after death.
- *9 The confession of faith serves as a reminder of the Day of Judgment.
- *10 A personal relation with God includes a proper recognition of his prophets.
- *11 I have faith in God as the Judge of mankind on the Day of Judgment.
- *12 All human relations shall be subjected to careful scrutiny on the Day of Judgment.
- *13 God shall mete out justice on the acts of all people after death.
- 14 I believe in life hereafter.
- 15 The nature of one's moral life shall be determined by one's conduct in life.
- 16 Hardly a day goes by without me judging my actions as well as the action of others.
- 17 I worry about bad things happening to good people.
- 18 I would prefer a good life over along one that is not good.
- 19 My life is always being judged by others as I judge theirs.
- 20 I worry that I tend to think that I am the measure of all things.

*Suggested items for a final scale.

ATTITUDES TOWARD FUNERALS

David Lester & Joshua Blustein
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Many attitudes toward death scales contain items related to death, the process of dying and funerals. Collett and Lester (1969) devised separate scales to measure the fear of death and dying, for oneself and for others, and the present study sought to devise a scale to measure attitudes toward funerals. Twelve items appeared to relate to three separate issues: attitudes toward the funeral industry (On the whole, I think that funeral industry does a pretty good job; The funeral industry is in need of much closer government inspection and control), viewing (Viewing the deceased in an open casket is an important part of the funeral; Cosmetic preparation of the deceased person is a fitting part of the funeral preparation; It is psychologically important for the close relatives of the deceased to view the body), and funerals (Funerals are times for family members to get close to one another; Funerals make me. anxious; People spend too much money on funerals; A funeral is an experience of great value; Funerals provide what the public wants; Funerals assist the emotional adjustment of those who have suffered the loss; A funeral can be a fitting memorial for someone who has died). These items were presented in a Likert-type format in a questionnaire.

The items were given to two samples: 49 students (Mage=21.6, SD=2.2) and 53 students (Mage=21.7, SD=3.5). Scores on the three subscales were significantly correlated in both samples (Pearson r s=0.32 to 0.52). The subscale scores did not correlate consistently with age (r s=-0.33 to -0.07), sex (r s=-0.05 to 0.12), belief in an external locus of control (r s=-0.03 to 0.08), fear of death of self (r s=-0.06 to 0.11), death of others (r s=-0.10 to -0.06), dying of self (r s=-0.18 to 0.17), or dying of others (r s=0.01 to 0.23). Thus, data from the present study indicate that attitudes toward funerals constitute an independent component of attitudes toward death and dying.

References

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We are interested in your views about funerals. For each of the items, please indicate how strongly you agree or disagree using the following scale. Please answer every item.

- +3strongagreement
- +2moderateagreement
- +1slightagreement
- 1slightdisagreement
- 2moderatedisagreement
- 3strongdisagreement

1. The funeral industry is in need of much closer government inspection and control
2. It is psychologically important for the close relatives of the deceased to view the body
3. Funerals are times for family members to get close to one another
4. Funerals make me anxious.
5. Cosmetic preparation of the deceased person is a fitting part of the funeral preparation
6. People spend too much money on funerals
7. A funeral is an experience of great value
8. On the whole, I think that the funeral industry does a pretty good job
9. Funerals provide what the public wants
10. Funerals assist the emotional adjustment of those who have suffered the loss
11. Viewing the deceased in an open casket is an important part of the funeral
12. A funeral can be a fitting memorial for someone who has died

AN INVENTORY TO MEASURE HELPLESSNESS, HOPELESSNESS, AND HAPLESSNESS

David Lester

Beck, Weissman, Lester, and Trexler (1974) published a 21-item scale which they proposed as a measure of "pessimism" but which has become known as a measure of "hopelessness." Since its appearance, the scale has been used in several hundred research studies. It has been a strong correlate and predictor of suicidality, stronger in many cases than measures of depression (see Farran, et al., 1995).

Perusal of the items included in the Hopelessness scale, however, indicate that some appear to tap helplessness rather than hopelessness. For example, "I might as well give up because I can't make things better for myself" seems to assess helplessness, whereas "My future seems dark to me" seems to assess hopelessness.

Lester (1998) devised three subscales of 10 items each to measure helplessness and hopelessness and, in addition, haplessness (the state of having bad luck or bad fortune), using the ideas contained in the pessimism scale (Beck, et al., 1974) and the belief in locus of control scales devised by Rotter (1966) and Levenson (1974). The items were modified for the present inventory and placed in Likert-type format rather than the yes-no formats of the pessimism and locus of control scales. The items are shown in the Appendix (p. 498).

Lester (1998), for a sample of 141 college students, found that scores on the three subscales were associated with scores on the Beck Depression Inventory (Beck, et al., 1961): for Helplessness Pearson $r = 0.37$ (two-tailed $p < .001$), for Hopelessness 0.43 ($p < .001$), and for Haplessness 0.21 ($p < .05$).

In a re-analysis of this data set, the Cronbach alphas of reliability were for Helplessness 0.79 , for Hopelessness 0.74 , and for Haplessness 0.79 . A factor analysis using a principal components extraction and a varimax rotation identified eight orthogonal factors which did not match the three subscales. By eliminating items with small loadings on the factors, three 4-item subscales were devised whose factor-analysis pattern matched the three subscales. The three values of Cronbach alpha were for Helplessness 0.63 , for Hopelessness 0.65 , and for Haplessness 0.69 . These brief subscales are indicated in the Appendix.

The present study was designed to explore the correlates of these measures of helplessness, hopelessness, and haplessness further and to check upon the reliability of the scales. A questionnaire was constructed containing the measures of helplessness,

hopelessness, and haplessness, along with measures of manic and depressive tendencies (Thalbourne, et al., 1994) and a questionnaire on the ownership of life (Ross & Kaplan, 1993-1994) which assesses the extent to which people believe that God, the government, or the individual has the power to control and govern one's life.. The questionnaire was administered to 19 male and 50 female undergraduate students enrolled in social science courses at a state college. Their mean age was 23.1 yr. (SD: 5.8).

The mean scores (and standard deviations) were for Helplessness 25.5 (SD=8.3), for Hopelessness 20.5 (SD=6.1), for Haplessness 27.2 (SD= 6.3), for mania 5.3 (SD: 1.5), for depression 4.0 (SD: 2.0), for trust in God 21.1 (SD: 7.4), for trust in the government 26.9 (SD: 4.0), and for trust in oneself 15.5 (SD: 4.3).

The Cronbach alphas for the scales were as high as those reported by Lester (1998): Helplessness 0.87, Hopelessness 0.80, and Haplessness 0.75. Thus, cross-validation of the scales was not associated with a decrease in reliability. Helplessness scores were significantly associated with depression scores and trust in the government scores (Pearson $r_s = 0.47$, two-tailed $p < .001$, and -0.24 , $p < .05$, respectively),. Hopelessness scores were significantly associated with mania scores, depression scores, and trust in God scores [$r_s = 0.25$ ($p < .05$), 0.50 ($p < .001$), and -0.31 ($p < .011$, respectively)]. Haplessness scores were not associated with any of the other measures.

The present study has indicated that reliable scales can be developed to measure helplessness, hopelessness, and haplessness. Although scores on the original 10-item scales were significantly associated, briefer 4-item scales were devised that were independent in a factor analysis. Looking at correlates of the scores for Helplessness, Hopelessness, and Haplessness, Hopelessness was the strongest correlate of depression and mania scores. Scores on Hopelessness were associated with placing less trust in God, whereas scores on Helplessness were associated with placing less trust in the government, results which support the construct validity of the scales. It is hoped that these scales will stimulate further research on the determinants and correlates of helplessness, hopelessness, and haplessness.

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APPENDIX

Helplessness

1. I can do just about anything I set my mind to.
4. I certainly feel useless at times.
7. I don't seem to be able to cope with crises without the help of others.*
10. I can hardly ever find ways around the problems that I face.
13. When I find myself in a jam, I can never think of ways of getting out of it.*
16. I have difficulty starting to do things.*
19. I rarely feel in control of my life.
22. Sometimes I think I may as well give up because there's nothing I can do about making things better for myself.
25. There are few ways around the problems that I am facing now.
28. I can't think of reasonable ways to reach my current goal.*

Hopelessness

2. I am confident that I will complete college.*
5. I look forward to the future with hope and enthusiasm.*
8. I don't expect to get what I really want.
11. I have enough time to accomplish the things I most want to do.*
14. In the future I expect to succeed in what concerns me most.*
17. All I can see ahead of me is unpleasantness rather than pleasantness.
20. When I look ahead to the future I expect I will be happier than I am now.
23. It is very unlikely that I will get any real satisfaction in the Future.
26. I can look forward to more good times than bad times.
29. I never get what I want, so it's foolish to want anything.

Haplessness

3. Many of the unhappy things in my life are partly due to bad luck.*
6. Trusting to fate has usually turned out well for me.
9. Many times, I might just as well decide what to do by flipping a coin.*
12. I have often found that what is going to happen will happen.

- 15. When I get what I want, it's usually because I'm lucky.
- 18. It's mainly a matter of fate whether or not I have a few friends or many friends.
- 21. To a great extent, my life is controlled by accidental happenings.
- 24. Often there's no chance of protecting my personal interest from bad happenings.*
- 27. It's not always wise for me to plan too far ahead because many things turn out to be a matter of good or bad for me.*
- 30. When I fail at things, I find it's useless to try again because my luck never changes.

*The items in the brief subscales.

AN ONTOLOGICAL INSECURITY SCALE

David Lester & Jeffrey Thinschmidt

R. D. Laing (1967, 1969) proposed that a psychologically healthy person has a sense of presence in the world as a real, alive, whole, and temporally continuous person. Two common splits to this holist self can occur. First the mind can be felt as disembodied, with no sense of being biologically alive. Second, the real self can erect a false pseudo self behind which to hide from others.

Laing presented the case of Mrs. R. who never convinced herself that she was somebody. She was frightened lest she look in the mirror and see no one there. She had to be with people who knew her. In crowds where she was not known to anyone, she panicked. Her greatest joy was taking care of her widowed father (at least he noticed her) and in being the model for her lover, a sculptor.

The present study tried to operationally measure Laing's concept of ontological insecurity. Analysis of his writings suggested four components: a doubt that one really exists, a desire to be noticed and to be the center of attention, a desire to live on in people's memory after one's death and a belief that the soul or mind is not firmly rooted in the body. Items were written for each of these components and scores on the four scales correlated with measures of extraversion and neuroticism.

Method

The 24-item ontological insecurity scale was administered along with the Eysenck Personality Inventory (Eysenck & Eysenck, 1973) to 31 male and 36 female college students enrolled in courses (mean age = 24.4, SD = 8.4). The ontological insecurity items were presented with a 5-point answering scale ranging from very true for me (5 points) to not at all true for me (1 point). The items are shown in the Appendix.

Results

A factor analysis using SPSSX, Principal Component extraction with a varimax rotation, identified three factors (see Table 1). Factor I accounted for 32% of the variance and had loadings from seeking attention (0.83) and extraversion (0.84). Factor II accounted for another 22% of the variance and had loadings from doubting one's existence (0.83) and neuroticism (0.81). Factor III accounted for a further 19% of the variance and had loadings from the desire to remain in people's memory (0.72) and a belief in the ease of separation of the mind and the body (0.87),

A doubt in one's own existence was significantly correlated with the desire to be the center of attention (Pearson $r = 0.25$, $p < 0.05$), a belief that the mind and the body are easily separated ($r = 0.38$, $p < .09$), and neuroticism scores ($r = 0.36$, $p < 0.001$).

Discussion

The present study operationalized four possible dimensions of Laing's concept of ontological insecurity. It was found that the four components did not correlate very highly with one another and, in a factor analysis with two personality scores, formed three factors.

Table I: Factor analysis of the components of ontological insecurity and extraversion/neuroticism

	Factor		
	I	II	III
Mind/body separation	-0.10	+0.23	+0.87#
To live on in memory	+0.35	-0.16	+0.72#
Doubts about existence	+0.04	+0.83#	+0.27
Desire to be noticed	+0.83#	+0.22	+0.10
Neuroticism	+0.09	+0.81#	-0.14
<u>Extraversion</u>	<u>+0.84#</u>	<u>-0.05</u>	<u>+0.04</u>
Percent of variance	32.0%	22.3%	19.2%
# high loading			

Doubts in one's own existence were found to be related to a belief in the ease of separation of mind and body and to neuroticism scores. This suggests that doubts in one's own existence can be a component of neurotic anxiety.

The present study suggests that Laing's concept, of ontological insecurity can be measured and that those who are ontologically insecure may have higher levels of neuroticism.

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Appendix

Doubts about one's existence:

1. I have occasionally wondered whether I really exist.
2. I have wondered whether I am part of someone else's dream, and when he/she wakes up I will no longer exist.
3. I used to wonder sometimes whether I would go to sleep and never wake up again.
4. It is incomprehensible to me that I could one day stop existing.
5. I have wondered whether I might look in a mirror and not see my reflection.
6. I have always wanted to be a model for an artist, sculptor, or photographer.

The desire to be noticed:

1. When I'm in a large crowd, I like to be noticed.
2. I like being the center of attention.
3. I avoid catching sight of myself in the mirror.
4. I like to be late to social gatherings so that I can be noticed when I arrive.
5. I am concerned with my physical appearance.
6. I don't feel that I am really complete unless I am involved with someone romantically.

The desire to be remembered after death:

1. It is important to me that my family name be carried on.
2. I will live on through my children and their children.
3. I would like to leave my mark on the world so that people will remember me.
4. If I died today, people would soon forget me.
5. I usually write my name on things I own (like books, etc) or create (like drawings, etc).
6. When I die I want to be buried and have a headstone for my grave.

Mind and body separation:

1. I believe in life after death.
2. I believe in reincarnation, that is, that we all have had previous lives and will have future lives.
3. Sometimes I have felt that my spirit or soul has left my body.
4. I am not concerned too much with my body since it is my mind, spirit, or soul which is the essence of my existence.
5. I believe that there is a heaven.
6. I believe that the mind has the power to cure bodily illnesses.

THE HOLISTIC THEORY OF ANDRAS ANGYAL: MEASURING THE SYSTEM PRINCIPLE

David Lester & Benjamin Dench

In Andras Angyal's (1965) holistic theory of personality, the mind is organized by a two-part system principle. The biopositive (or healthy) system principle is composed of two parts; a trend toward autonomy and a trend toward homonomy. The trend toward autonomy involves self-expansion at the expense of the environment and includes the desires to explore, achieve, dominate, defend one's integrity and acquire possessions. The trend toward homonomy involves the desire to integrate with others, such as a family, group or culture, and to feel at one with Nature, God or the universe.

Angyal proposed two patterns for the bionegative (or neurotic) system principle. The method of adjustment in the 'pattern of vicarious living' is the systematic suppression of one's genuine personality and an attempt to replace it with a façade self. This is the self one presents to others and by means of which one hides the real self. This pattern develops because the person felt unloved and disliked as a child. To receive love and approval, they suppress their real selves and become what they think others want them to be. This pattern leads to the hysterical personality and, when negativism is present, the antisocial personality.

The 'pattern of noncommitment' characterizes the person who is uncertain whether the world is basically good and friendly or bad and hostile, and so who expresses uncertainty and ambivalence. Such a pattern arises when children face an inconsistent world: today the parents are kind and friendly, but yesterday they were angry. This pattern leads to anxiety and a search for ways to dispel the anxiety, frequently resulting in the obsessive and compulsive personality.

Dench, Martino and Lester (2008) noted that autonomy is opposite to the pattern of vicarious living, a contrast between satisfying the self versus suppressing the self and adopting a façade self. Similarly, homonomy is opposite to the pattern of noncommitment, that is, a moving toward others versus moving away from others. The present study sought to operationalize these two trends and two patterns, using simple self-report scales, and to explore some psychopathological correlates of the trends and patterns.

Table 1: Pearson correlations between the scale scores

	Autonomy	Homonomy	Non-commitment	Vicarious living	Mean score (SD)
Autonomy	-	+0.51***	-0.18	-0.32**	6.37 (1.23)
Homonomy		-	-0.22 ²	-0.15	5.36 (1.45)
Non-commitment			-	0.44***	3.47 (1.67)
Vicarious living				-	2.92 (1.81)
Psychoticism	-0.42***	-0.39***	-0.02	+0.04	
Neuroticism	+0.10	+0.06	+0.49***	+0.42***	
Extraversion	+0.16	+0.19	-0.07	-0.08	
Lie	-0.07	+0.18	-0.19	-0.12	

* two-tailed $p < .05$

** two-tailed $p < .01$

*** two-tailed $p < .001$

Seven questions were written to capture the essence of each of trend (autonomy and homonomy) and each pattern (vicarious living and noncommitment); see the Appendix below. They were written for this study or modified from prior scales, including the Plural Self Scale (Altrocchi, 1999), a scale to measure Maslow's Hierarchy of Needs (Lester, 1990) and the Sociotropy-Autonomy Scale (Bieling, Beck & Brown, 2000). They were presented with an answer format of agree versus disagree. The questionnaire was administered anonymously to 62 female and 20 male undergraduates enrolled in psychology courses at a rural state college ($M_{age} = 21.5$ yr., $SD = 5.2$), along with the Eysenck Personality Inventory (Eysenck, Eysenck & Barrett, 1985) which measures the personality traits of Psychoticism, Neuroticism and Extraversion. Mean scores for the four scales are shown in Table 1. Cronbach alpha reliabilities were: autonomy 0.72, homonomy 0.58, non-commitment 0.49 and vicarious living 0.60. ranged from 0.49 to 0.73 (see Table 1), modest at best.

As predicted, scores for Autonomy and Vicarious Living were negatively associated (see Table 1), as were scores for Homonomy and Noncommitment. Scores for Autonomy and Homonomy were negatively associated with Psychoticism scores (see Table 1) while scores for Vicarious Living and Noncommitment were positively associated with Neuroticism scores.

The results confirm a negative association between Autonomy and Vicarious Living, and a negative association between Homonomy and was Noncommitment. Respondents with higher biopositive scores (Autonomy and Homonomy) had lower Psychoticism scores, while respondents with higher bionegative scores (Vicarious Living and Noncommitment) had higher Neuroticism scores, confirming the biopositive nature of the trends and the bionegative nature of the patterns. These conclusions are limited by

² Two-tailed $p = .059$

the small sample size, the small proportion of men, and the low reliability of the scales, but the support for the predictions of the study suggest that the hypotheses merit further study.

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The Questions

Autonomy

If a goal is important to me, I will pursue it
 I set my own standards and goals for myself
 When I achieve a goal, I get more satisfaction from reaching the goal than from any praise I might get
 It is important to me to meet my own objectives for a task
 I enjoy accomplishing things
 It is important that I myself know that I've done a good job.
 It is important to me to satisfy my own needs

Homonomy

I often find myself thinking about my friends and family
 I like to spend my free time with others
 I need to be close with my family
 My roots, cultural and ethnic, are important to me
 I like to have a few intimate friends that I can rely on and who can rely on me
 I sometimes feel one with God or the universe
 I get pleasure from helping others

Noncommitment

I think that world is a rather unsafe place these days
 I worry about whether or not I can trust other people
 I "touch on wood" and do other little acts to avoid bad things happening to me
 I find it difficult to predict how other people are going to react to me

I have little rituals that I think might bring me good luck during the day
I have persistent feelings of insecurity
I worry a lot about catastrophes like fires, earthquakes and airplane crashes

Vicarious Living

It is important to me to be liked and approved of by others
I censor what I say because I am concerned that the other person may disapprove or disagree
I worry that, if people knew my faults or weaknesses, they would not like me
I spend a lot of time of my appearance so that others will compliment me on how I look
Sometimes I am afraid I will be discovered for who I really am
I sometimes feel like a phony
My public and private self are the same person

MEASURING MASLOW'S HIERARCHY OF NEEDS

David Lester

I recently published an article on my inventory to measure Abraham Maslow's hierarchy of needs, and so I will not reprint that article here. However, I present the inventory below. The physiological needs are items 1, 6, 11, 16, 21, 26, 31, 36, 41 and 46, etc.

Lester, D. (2025). Measuring Maslow's hierarchy of needs. *Assessment & Development Matters*, 17(4), 18-21.

Here is a series of general statements. Indicate how much you agree or disagree with them. Put your answer in the blank space in front of each item according to the following scale:

+1 slight agreement	-1 slight disagreement
+2 moderate agreement	-2 moderate disagreement
+3 strong agreement	-3 strong disagreement

Read each item and decide quickly how you feel about it. Put down your first impressions. Please answer every item.

1. I never have trouble getting to sleep at night.
2. I think that the world is a pretty safe place these days.
3. I know my family will support me and be on my side no matter what.
4. I feel dissatisfied with myself much of the time,
5. I have a good idea of what I want to do with my life.
6. I have an income that is adequate to satisfy my needs.
7. I would not walk alone in my neighborhood at night.
8. I am involved in a significant love relationship with another.
9. I feel respected by my peers.
10. My life has meaning.
11. I get an adequate amount of rest.
12. My anxiety level is high.
13. I feel rootless.
14. I seldom have fears that my actions will cause my friends to have a low opinion of me.
15. I am uncertain about my goals in life.
16. I have a satisfactory sex life.
17. I feel secure about the amount of money I have and earn.

18. I have a group of friends with whom I do things.
19. I can stand on my own two feet,
20. I feel I am living up to my potential.
21. In general, my health is good.
22. I feel safe and secure.
23. I feel somewhat socially isolated.
24. I feel confident in my present field of endeavor.
25. I am seeking maturity.
26. In Winter, I always feel too cold.
27. I am afraid to stay in my house/apartment alone at night.
28. I have a few intimate friends on whom I can rely.
29. I would describe myself as a self-confident person.
30. I find my work challenging.
31. I eat enough to satisfy my physiological needs.
32. My life is orderly and well-defined.
33. I feel close to my relatives.
34. I have earned the respect of other s.
35. I know what my capabilities are and what I can't do.
36. I get an adequate amount of exercise.
37. I can depend on others to help me when I am in need.
38. I am interested in my ethnic roots and feel a kinship with others in my ethnic group.
39. I don't spend much time worrying about what people think of me.
40. I feel I am doing the best I am capable of.
41. There's usually some part of my body that is giving me trouble.
42. I am often worried about my physical health.
43. I am religious and consider myself to be a member of a religious group.
44. I feel that I am a worthy person.
45. I feel that I am growing as a person.
46. The summers are too hot for me to ever feel comfortable.
47. My life has a nice routine to it.
48. I am able to confide my innermost thoughts and feelings to at least one close and intimate friend.
49. In groups, I usually feel that my opinions are inferior to those of other people.
50. My educational achievements are appropriate given my ability.

ATTITUDES TOWARD LIFE AND AGING

David Lester, Linda Monfredo & Holly Hummel

Research has yielded few differences in fear of death by the young and the old, though the elderly do report unhappiness more often than the young (Veroff & Depner, 1978). To explore attitudes toward life and aging, two exploratory Likert-type scales were constructed: an 11-item scale to assess attitudes toward aging and a 13-item scale to assess attitudes toward one's current life. The attitude toward aging scale was completed anonymously by 64 senior citizens in the community ($M_{age} = 69.8$, $SD = 5.9$) and 71 college students ($M_{age} = 27.6$ yr., $SD = 9.6$). The attitudes toward current life scale was given to 36 senior citizens in the community ($M_{age} = 66.3$, $SD = 4.4$) and 52 college students ($M_{age} = 25.2$, $SD = 9.8$).

The senior citizens reported significantly less fear of aging than the young adults (mean scores —7.4 and 1.2, $SD = 11.5$ and 10.0 ; $t = 4.57$, $df = 134$, two-tailed $p < .001$). The two groups did not differ in scores on attitudes toward their current life (mean scores 24.8 and 21.8, $SD = 8.7$ and 7.9 ; $t = 1.42$, $df = 61$).

The scores of the groups on these scales were unrelated to sex, age (within each group), and educational level (some college education versus none for the elderly, years of college for the young adults). For the college students, scores on the scale assessing current attitudes toward life were negatively correlated with scores on fear of death (Lester, 1967) ($r = -0.48$, $df = 46$, two-tailed $p < .001$). Students with positive attitudes toward their current life had less fear of death. The present data suggest that senior citizens and college students are equally positive about their current life, but the senior citizens have much less fear of aging.

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The scales were administered in a Likert-type format.

Fear of Aging

1 I would experience a great loss if someone close to me died.

- 2 The total isolation of aging frightens me.
- 3 I am disturbed by the physical degeneration involved in aging.
- 4 I would not mind visiting a senile friend.
- 5 If I had a choice as to whether or not a close relative should be informed he/she is dying, I would tell him/her.
- 6 The pain involved in aging frightens me.
- 7 I am disturbed by the shortness of life.
- 8 The feeling that I might be missing out on so much after I die bothers me.
- 9 If I had a fatal disease, I would like to be told.
- 10 The intellectual degeneration of old age disturbs me.
- 11 I am disturbed by the thought that my abilities will be limited while I grow old.

Attitude toward Life

- 1 My life is important and meaningful for me.
- 2 As I grow older, the quality of my life seems to be getting better.
- 3 Life can be rewarding, even for a person who is not physically healthy.
- 4 I find making new friends to be difficult.
- 5 I enjoy being my age and wouldn't want to be any other age even if I could.
- 6 I am looking forward to doing some interesting and fun things in the future.
- 7 Most of my everyday activities tend to be boring and monotonous.
- 8 This is the most depressing time of my life.
- 9 The life situation of the average person seems to be improving.
- 10 My present living arrangements are unsatisfactory.
- 11 I am able to lead a good life even though I don't have as much money as I'd like to have.
- 12 I believe that my personality is attractive.
- 13 So far, I have been very lucky in life.

A SHORT COMPUTER ANXIETY SCALE

David Lester, Bijou Yang & Simon James

There have been many studies of the structure, correlates and sources of computer anxiety. For example, Beckers and Schmidt (2001) identified six factors of computer anxiety: computer literacy, self-efficacy, physical arousal associated with computer use, affective feelings about computers, beliefs about the beneficial effects of computers, and beliefs about their dehumanizing aspects. Beckers and Schmidt examined the pathways between these aspects and concluded that enhancing self-efficacy and computer literacy would reduce computer anxiety.

Such studies are excellent for exploring the complexity, correlates, and sources of computer anxiety. But when computer anxiety is but one predictor variable in a study of some other target variable, such as shopping online, measuring devices with large numbers of items and complex subscale structure are problematic for they increase the length of surveys and provide multiple measures of computer attitudes whereas what is required is a single measure of computer anxiety.

To remedy this and provide a simple quick measure of computer anxiety, a 6-item scale was devised and examined for its reliability and validity.

The Scale

A 6-item scale was devised with items which concerned confidence in using computers. Two items reflect comfort with computers and four inadequacy. The items require a Likert-type response, and the scale is shown in the Appendix. Reliability The scale was administered to 48 American students (9 men and 39 women, Ma., 22.1 yr., SD=4.8) with a 3-wk. test-retest interval. The Pearson correlation for scores on two occasions was .87 (two-tailed $p < .001$). The scale was administered to two samples of American students (35 men and 102 women, Mage=22.8 yr., SD=5.4; and 146 men and 219 women, Mage = 22.3 yr., SD = 4.4). Cronbach alpha for the scale was 0.78 for both samples. The items were subjected to factor analysis using a principal components extraction. Only one eigenvalue was greater than one, and the single factor extracted accounted for 49.6% and 49.8% of the variance in the two samples, respectively.

Table 1:

Table 1: Three factor analyses of the scale

Item	137 American Students	Sample 365 American students	142 English students
1	.68	.71	.66
2	.73	.78	.67

3	.64	.64	.62
4	.61	.63	.63
5	.80	.83	.79
6	.75	.61	.73
% of variance	49.6	49.8	47.5

The scale was also administered to 142 British students (77 men and 65 women, Mage=20.2 yr., SD= 1.9). Cronbach alpha for that sample was 0.76. A factor analysis identified a single factor with an eigenvalue greater than one, accounting for 47.5% of the variance. Means for the three large samples were 14.5 (SD=5.0) for 137 American students, 13.4 (SD = 5.1) for 365 American students, and 14.2 (SD = 4.5) for 142 British students.

Construct Validity

(1) The scale was administered to 71 American students (15 men and 56 women, Mage =21.8 yr., SD = 4.3). Computer anxiety scores were negatively associated with the total number of 20 computer and Internet activities ($v = -0.45$, $p < .001$), with the number of hours online ($r = -0.35$, $p = .004$), and with subscribing to an Internet Service Provider ($r = -0.28$, $p = .02$). Computer anxiety scores were positively but weakly associated with age ($r = 0.27$, two- tailed $p = .03$) but not sex ($r = -0.03$).

With respect to the individual items concerning computer and Internet behaviors, computer anxiety scores were significantly associated with six items. Those persons with higher computer anxiety scores were less likely to have written a program, installed extra memory, installed an extra disc drive, e-mailed photographs, or sent static or moving greeting cards.

(2) The scale was administered to 365 American students (146 men and 219 women; Mage, = 22.3 yr., SD = 4.4). Computer anxiety scores were negatively and weakly associated with hours spent online ($r = -0.24$, $p < .001$), ever purchasing something online ($r = -0.17$, $p = .001$), and having a higher score on a measure (Rotter, 1966) of belief in an external locus of control ($r = 0.17$, $p = .001$).

Of 15 items which can be purchased online, those participants with higher computer anxiety scores were less likely to purchase books, compact discs, computers, airline tickets, hotel rooms, stocks/shares, and pornography. They did not differ in purchasing cars, car insurance, stationery, groceries, clothes, shoes, shows/concert seats, or gambling. Computer anxiety scores were not associated with age ($r = 0.06$), but women scored somewhat higher ($r = 0.24$, $p < .001$).

(3) The scale was also administered to 142 British students (77 men and 65 women, Mage=20.2 yr., SD= 1.9). Those with higher computer anxiety scores had higher scores on the measure of belief in an external locus of control ($r = 0.30$, $p < .001$) and tended to spend fewer hours online ($r = -0.15$, $p = .09$). Although those with higher computer anxiety scores were not less likely to shop online ($r = -0.08$), they were less

likely to purchase books, shows/ concert seats, and gambling online. Computer anxiety scores were not associated with sex ($r = -0.14$) or age ($r = 0.13$).

Concurrent Validity

(1) The scale was administered to 137 American students (35 men and 102 women, $M_{age}=22.8$ yr., $SD=5.4$). Computer anxiety scores were associated with having lower scores on measures devised by Liaw (2002) of attitudes toward computers and attitudes toward the Internet ($r = -0.62$, $p < .001$, and $r = -0.39$, $p < .001$, respectively). As in the other groups, computer anxiety scores were also negatively associated with the total number of computer/Internet activities ($r = -0.46$, $p < .001$).

(2) In a sample of 52 American students (9 men and 43 women; $M_{age}= 22.3$ yr., $SD=5.0$), scores were significantly correlated with scores on Bear, Richards, and Lancaster's Computer Attitude Scale (1987) ($r = 0.38$, $p = .007$).

The reliabilities and validities were apparent in both American and English samples, providing evidence for cross-national usefulness of this little scale, which appears to be suitable as a quick measure of computer-related anxiety in research.

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APPENDIX

For each item, indicate your opinion by circling one answer:

SA: strongly agree

A: agree

MA: mildly agree

MD: mildly disagree

D: disagree

SD: strongly disagree

1. I feel confident and relaxed while working on a computer*
2. The harder I work at learning computers the more confused I get.
3. I have sometimes thought that I am too old to learn about computers
4. I have sometimes thought "Computers don't like me."
5. I always have problems working on computers
6. I can usually manage to solve computer problems by myself*

* reverse scored.

AVERSION TO TOUCHING

David Lester & Alisha Rencher

Physical touch can communicate many meanings to others ranging from affection to dominance. In experimental workshops, exercises which involve touching one another are often used to break down the barriers between people. For example, a person with eyes closed may explore the face and body of a partner.

While some people enjoy and may learn from these experiences, many people are made uncomfortable by having to touch others. The present study was designed to explore the attitudes of people toward touching and whether these attitudes are related to personality.

Method

A questionnaire of 13 items was constructed to measure attitudes toward touching (see Table 1). These were placed in a Likert-type format and administered to 12 male and 35 female students under the age of 30 (mean age=20.7, SD=2.1) enrolled in undergraduate courses. The students also completed the short form of the Eysenck Personality Inventory (Eysenck, et al., 1985). Items measuring whether touching was viewed as flirtatious, showing concern or showing anger were included in a preliminary study but found to be unrelated to personality scores.

Results and Discussion

Responses to the touching attitudes scale were subjected to a factor analysis using a principal components extraction and a varimax rotation, and two factors were identified (see Table 1). Factor I appeared to measure whether the respondent touches others or not (behavior) while Factor II appeared to measure whether touching makes the respondent feel anxious or uncomfortable (emotion).

Scores for the behavior component (Factor I) were significantly associated with extraversion scores, with the more extraverted respondents touching others more (see Table 1). Factor scores for the emotion component (Factor II) were significantly associated only with psychoticism scores, with those feeling more comfortable with touching scoring higher for psychoticism.

Table 1: Results of the factor analysis and correlations with personality

Item	Factor I	Factor II
1 When I talk to a person, I usually touch them	+0.75#	+0.22
2 I like to be touched	+0.26	+0.51#
3 It makes me feel uncomfortable to touch someone while talking	-0.59*	-0.39

4 I feel uncomfortable being touched by someone I don't know	-0.05	-0.74#
5 Being touched by someone makes me feel more secure when talking to that person	+0.48	+0.64#
6 I do not like it when people come too close to me	-0.17	-0.75#
7 Physical closeness with a stranger does not bother me	-0.04	+0.84#
8 I often touch a person's hand while talking	+0.67#	+0.32
9 I never touch people while talking to them	-0.71#	-0.16
10 Touching an arm or hand while talking is not necessary	-0.73#	-0.02
11 I sometimes touch someone to let them know that I am interested in their conversation	+0.77#	+0.12
12 Touching shows feeling and understanding while conversing with someone	+0.59#	+0.45
13 If a friend touched me while talking, I would think they were strange	-0.67#	+0.02
Percent of variance	40.9%	14.0%
Pearson correlation with:		
Psychoticism	+0.01	+0.32*
Extraversion	+0.32**	+0.17
Neuroticism	+0.21	-0.12
Lie	-0.08	+0.10
# Loading >0.50	*p< .02	

These associations were found even after controls (by means of partial correlations) for age and sex. The association between the behavior of touching and extraversion was found in both men and women, while the association between psychoticism and feelings was found only in men.

The association between touching scores and extraversion was not surprising. Extraverts, commonly described as sensation-seekers, would be expected to touch others more. In contrast, the association between psychoticism and feeling *more* uncomfortable with touching is unexpected. However, the respondents did not obtain high psychoticism scores – the mean score was 2.7 (SD=1.5) – and this association may not be replicable using clinical samples whose psychoticism scores would be much higher.

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THE DESIRE TO LOSE ONE'S IDENTITY

David Lester

Laing (1969) proposed that a psychologically person has a sense of presence in the world as a real, alive, whole person rooted in a physiological body. Recently, Lester and Thinschmidt (1988) devised a scale to measure this belief. And found that doubts about one's existence were associated with the level of neuroticism

Maslow's (1970) discussion of the concept of the self-actualized person suggests, in contrast, that some individuals, while having a secure sense of self, occasionally seek to lose their identity. This may be achieved in various ways. Some individuals inhibit their own desires to achieve and obtain gratification from the achievements of those close to them (for example, a spouse or a child). Others have the fantasy of merging identities with their lover, so that the two become one.

The present study sought to devise a scale to measure these desires and to explore with what personality traits they were associated. Items were written to tap five areas: aversion to physical closeness with another, resentment of others having an influence over you, aversion to merging minds with another, disliking identification with the achievements of others, and unsociability. Eight items were written to tape each of these five areas.

Method

A questionnaire containing the items was administered to 36 men and 48 women aged 18-25 (mean age 21.1 years, SD=1.6) enrolled in undergraduate courses. The items were placed in a Likert-type format, scored for aversion. The students also completed the Eysenck Personality Inventory (Eysenck, Eysenck & Barrett, 1985) which measures neuroticism, extraversion and psychoticism.

Results and Discussion

Cronbach's alphas for the five scales were: aversion to physical closeness 0.54; resenting the influence of others 0.46; aversion to merging with others 0.70; aversion to identification with the achievements of others 0.55; and unsociability 0.57.

The dimensions measured by the scales did not appear to be pathological (see Table 1). Psychoticism scores correlated with aversion to physical closeness (Pearson $r = 0.25$, $p < 0.05$), aversion to merging ($r = 0.26$, $p < 0.05$), and aversion to identification with the achievements of others ($r = 0.30$, $p < 0.01$), but not significantly with resentment of others having an influence over oneself or with unsociability ($r_s = 0.01$ and -0.20).

The aversion to merging with others was negatively associated with neuroticism scores ($r = -0.22$, $p < 0.05$), indicating that those with higher neuroticism scores had less aversion to merging with others. As expected, extraversion was negatively correlated with unsociability ($r = -0.70$, $p < 0.001$).

To explore the correlations between the five dimensions, the scores were subjected to a factor analysis using SPSSX, with a principal components extraction and a varimax rotation. Two factors were extracted (see Table 1), the first of which, accounting for 39.2% of the variance, with loadings from physical closeness, merging and identification with others, correlated significantly with psychoticism ($r = 0.39$, $p < 0.001$). Factor II, accounting for 21.1% of the variance, with loadings from physical closeness and unsociability, was associated with extraversion ($r = -0.56$, $p < 0.001$).

Table 1: The correlations between scale scores and results of the factor analysis

	P	E	N	L	Factor	
					I	II
Aversion to						
Physical closeness	+0.25*	-0.09	-0.11	-0.02	+0.55#	+0.57#
Influence from others	+0.01	+0.10	-0.17	+0.07	-0.30	-0.35
Merging with others	+0.26*	-0.06	-0.22"	-0.10	+0.81#	+0.23
Identification with others	+0.30**	-0.11	-0.06	-0.09	+0.81#	-0.11
<u>Sociability</u>	-0.20	-0.70**	+0.15	+0.20	-0.12	+0.89#
<u>Percent of variance</u>					39.2%	21.1%
Psychoticism (P)					+0.39**	-0.13
Extraversion (E)					+0.06	-0.56**
Neuroticism (N)					-0.18	+0.12
<u>Lie (L)</u>					-0.16	+0.15

high loading.

*two-tailed $p < 0.05$; **two-tailed $p < 0.01$.

I

It can be seen that the desire to merge identities with an intimate partner is not a severely pathological desire. Rather it is related to other social desires, such as the desire to be physically close to others and to identify with the achievements of others.

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APPENDIX

The Scale Items (Some items are phrased negatively)

Physical closeness

1. I avoid physical contact with casual acquaintances
2. I like to sleep close to my partner in a double bed
3. I hate to do things without my partner
4. I often hug my friends when I meet them after an absence
5. I feel-uncomfortable, when others embrace me
6. I like the feeling of being intertwined during lovemaking
7. When I'm upset I like to be hugged and held
8. I like holding hands when I walk with my partner

Others' influence

1. I resent being told by other people how I ought to feel
2. As a child I used to sometimes fear that my parents could read my mind
3. I enjoy being exactly who I am
4. Friends and family often accuse me of being stubborn
5. I hate changing myself in order to make my partner happy
6. I refuse to change my beliefs to please others
7. I am easily influenced by others
8. I resent it when others are trying to influence the way I think

Merging

1. I like it when the person I feel close to seems to know what I am going to say and can even finish my sentences for me
2. I can never let myself get really intimate with someone else
3. I have occasionally wished I could be inside my lover's mind and he/she in mine so that we would each know how the other feels
4. The ultimate in intimacy is when my partner understands my thoughts and feelings
5. I have never truly emotionally 'meshed' with anyone
6. I feel uneasy when my partner seems to understand me too well
7. It upsets me to feel that there is any barrier between me and my partner
8. I don't ever want to be really close to someone else.

Identification with others

1. It upsets me emotionally to see other people suffer
2. I derive a great deal of satisfaction from the successes of my partner
3. I enjoy my family's accomplishments as much as my own
4. I resent the amount of time demanded of me by my family obligations
5. I get emotionally upset by tragic situations elsewhere in the world

6. I would feel that my life was incomplete if I never had any children of my own
7. I sometimes resent it when a member of my family is successful at something
8. I resent it when I am introduced as someone else's brother or wife or friend etc.

Sociability

1. I get more enjoyment from my material possessions than from my relationships with others
2. I feel anxious in crowds
3. I feel uncomfortable at large parties
4. I would rather be in a group than spend time alone
5. I get a lot of satisfaction from my relationships with others
6. I find it easy to be the "life and soul" of the party
7. I like to be the center of attention of people
8. In groups of people I prefer to listen to and watch the others than to be very active

THE SHAW BLOCKS TEST³

David Lester

The Shaw Blocks Test has been the subject of much research (1-7) in recent years. However, the test is not available for sale to other psychologists who would care to use it. In order that others may construct sets of the test, this paper describes the blocks in detail.

The Shaw Blocks Test consists of four blocks. Each is a cylinder, but the cross-section of each is a different shape: triangle, square, pentagon, and hexagon. The blocks vary in weight, length of side, and depth. Each block has a hole cut into the center of the top face, varying in diameter and depth. On this top face there is a letter of the alphabet: A, B, C, and D. Into the side opposite this letter a notch is cut which again varies in length, depth, and position along the side of the block. Each block is painted with a different shade of gray. On the base of each block the name of an animal is printed: ox, rat, lamb, and whale. This varies in the size of print and the distance between the base of the name and the side of the block. This is summarized in Table 1, which gives the measurements of each of these features in millimeters.

The person is asked to inspect and pick up the blocks. When he has done this, he is asked to arrange them in a series and to explain his reason for choosing that particular order. He is required to give as many sequences as he is able in twelve minutes. Heim and Watts (4) have used four pieces of paper which vary in size, number of lines drawn on the piece of paper, and shade of the color with which the lines are drawn, in order that the subject may be shown examples of the required response.

Howson (5) and Bromley (2) have described methods of scoring responses to the test, but Lester has described a more adequate method, which does not penalize originality and which allows an analysis of the different types of response that are possible.'

The Shaw Blocks Test appears to be a measure of intelligence. Lester (6) found a high correlation between scores on the Shaw Blocks Test and scores on the Miller Analogies Test (8). Bromley (2) considered the test to measure creative intellectual output and measured the decline of this quality in aging adults. Howson (5) used the test to assess abstract thought and to differentiate brain-injured patients from neurotics and normal people. Lester (7) has described how a measure of originality can be obtained

³ I obtained permission from Harold Shaw back in the 1960s to publish this article on the blocks that he designed. They are shown in this website for the Science museum in London, England, , with the maker listed as unknown: <https://collection.sciencemuseumgroup.org.uk/objects/co8063439/shaw-psychological-test-united-kingdom-1948>

from the blocks which is only slightly correlated with the Lester point-score. (This correlation was not significantly different from zero.) Both scores can be obtained in one administration of the test.

Table 1: The dimensions of the Shaw blocks. All distances are in millimeters.

Letter on the block	A	B	C	D
Number of sides of top face	6	4	5	3
Shade	2	3	1	4(lightest)
Weight	2	4	3	1(heaviest)
Diameter of hole	31	12	25	19
Depth of hole	11	16	25	6
Length of side	31	51	37	61
Depth of block	48	31	38	44
Length of notch	6	25	19	12
Depth of notch	19	25	6	12
Breadth of notch#	6	6	6	6
Distance of left end of notch from left end of block side	16	26	0	18
Name of animal	Lamb	Rat	Ox	Whale
Height of print	7	5	15	11
Length of name	31	12	26	40
Distance from base of name to side of block	9	5	8	7
Side of block along which name is printed*	3	3	2	1

#Breadth indicates the distance through which the notch extends into the top face of the block
•See Figure I.

In Figure 1 (shown below), the face of the blocks shown is the top face, and the numbers are the sides of the blocks numbered for convenience of description in Table 1.

The advantages of the Shaw Blocks Test are many. It is open-ended and, if appropriately scored, does not penalize originality, as do many tests of intelligence. Since it is non-verbal it can be given to the deaf and the dumb and to foreigners. It can be given to illiterates for whom no suitable intelligence tests yet exist. It can be given to adults and children. The only requirement is possession of a mental age of about eight (Heim & Lester [3]). It can also differentiate among a highly selected population such as graduate students.

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Manual for the Shaw Blocks Test.

The administration of the Shaw Test

The subject is seated comfortably before a desk. The four blocks are placed before him in a random arrangement. He is told, "Will you have a look at these four blocks? Pick them up and handle them". After the subject has examined the blocks he is told, "You will see that they can be arranged in a logical order from left to right; in a sequence or series. There are a number of ways of arranging them. I want you to think of a reason for putting the blocks in a particular order, then to arrange them in that order, and when you have done so to tell me your reason for choosing that order. I want as many orders as you can think of in twelve minutes".

The subjects are given twelve minutes for the test. If they stop before twelve minutes has passed they should be encouraged to continue. The responses are noted in the order in which they are given. For each response, the position of the blocks in the sequence is noted. If the subject indicates that he does not understand the instructions, he should be given a demonstration with the blocks. He should be given examples of responses until he understands the problem.

The scoring system

In the Lester point-scoring system the responses are divided into five classes.

A responses are responses which are unique and objective, and which can be put in order simply by examining the blocks; for example, the length of the sides, and 'the distance between the edge of the hole and the edge of the notch'.

B responses are unique and objective, but need guess-work to order the blocks correctly; for example, the volume of the blocks and the frequency with which the animals names occur in the Bible'.

C responses are series which are subjective; for example, the attractiveness of the animal.

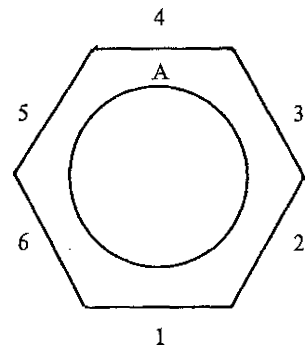
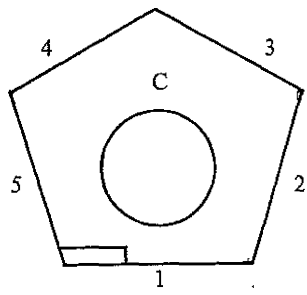
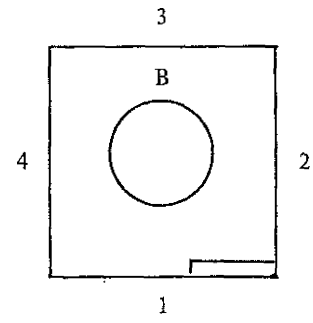
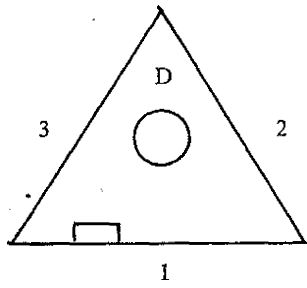
D responses are not series. They are groupings, nonsense sequences, and analogies; for example, 'these three animals live on land and this one in the sea'.

E responses lack all idea of coherence and consist of descriptions of the blocks and attempts to use them as toys; for example, 'all of them have holes'.

- A responses score 2 marks for the idea and 1 for the correct order of arrangement.
- B responses score 2 marks.
- C responses score 1 mark.
- D responses score -1 mark.
- E responses score -2 marks.
- Reversals: 1 mark is given for the first reversal and no marks for subsequent ones.
- If the principle is stated (that is, that all series can be reversed) 3 marks are scored.
- Alternations: These are scored in the same way that reversals are scored.
- Demonstrations: For each demonstration required and given 3 marks are deducted.
- Repetitions: If a response is restated no marks are given. If an equivalent response is given 1 mark is scored, while if its equivalence is noted by the subject 2 marks are scored; for example, number of corners and number of sides are equivalent responses.

Some of the problems that arise in the procedure are dealt with here.

- Subjects should not be encouraged to form series if they merely state possible reasons for sequences unless they have misunderstood the instructions. If this were done, it would have to be done for all responses and this would suggest to the subjects that D and E responses were wanted by the tester. Thus they would be encouraged to give them even though realizing that they were not logical series. Hence their scores would be adversely affected.
- If a verbal response is made and then either the subject says that it is 'stupid' or ignores it, then no score is given. This is to allow thinking aloud.
- If a subject asks, "Do you want reversals (or alternations)?" he is told no and credited with having stated the principle.
- If a subject asks whether groups are allowed, the instructions are re-read to him



The Shaw Blocks

THE EXECUTIVE PERSONAL FINANCE SCALE

Marcello Spinella, Bijou Yang & David Lester

Preliminary research has shown that self-rated measures of executive function relate to aspects of personal finance management (Spinella, et al., 2004a, 2004b). Credit card debt correlated positively with symptoms of prefrontal dysfunction in a community sample after controlling for demographic variables (age, sex, education and income). A strong association was found with the E (executive) scale of the FrSBe scale, which measures conceptualizing and organization ability. Furthermore, prefrontal dysfunction is related to delay discounting, and preferring smaller more immediate rewards over larger delayed rewards (Spinella, et al., 2004a, 2004b).

Given this evidence, the present study aimed to create a self-rating scale that specifically measured executive function aspects of personal finances.

Method

Participants

Participants were 225 adults (36% male), aged 18-64 years ($M = 26.2$, $SD = 10.3$) who had completed between 9-21 years of formal education ($M = 14.7$, $SD = 1.6$).

Measures

Personal Finances

Aspects of personal finances were obtained using a questionnaire. Participants reported personal gross annual earnings between \$0 and \$200,000 per year ($M = \$19,909$, $SD = \$26,450$). Current total credit card debt ranged from \$0 to \$27,000 ($M = \$1,534$, $SD = \$3,415$), and lifetime maximum credit card debt ranged from \$0 to \$40,000 ($M = \$2,588$, $SD = \$5,180$). A current debt-to-income ratio was calculated for each person using current credit card debt and personal gross annual income, which ranged from 0.0 to 0.9 ($M = 0.1$, $SD = 0.2$). Participants were also asked about their financial investment methods: 84% utilized a savings account,

28% utilized certificates of deposit, 21% utilized mutual funds, 26% invested in the stock market, 21% had a retirement account, and 13% invested in real estate.

Executive Personal Finance Scale

The Executive Personal Finance Scale (EPFS) is a novel scale that was created for this study. An initial pool of 26 items was created regarding executive functions as they pertain to finances. Items were created to reflect different domains of prefrontal system function, including conceptualization of finances, organization, financial planning, and impulse control over spending. These items were subjected to statistical analyses in order to construct a scale, as detailed later.

Money Attitudes Scale (MAS)

The MAS is a 29-item scale that was developed as a self-rating measure to assess attitudes toward money along four scales: Power-Prestige, Retention-Time, Distrust, and Anxiety (Yamauchi & Templer, 1982). Power-Prestige (MASpp) items relate to the tendency to use money as a means to impress others and measure success. Retention-Time items (MASrt) relate to the tendency to save money and plan financially. Distrust items (MASdt) relate to a hesitant attitude toward spending and suspicion that better prices for purchases could have been found elsewhere. The Anxiety subscale (MASanx) relates to various aspects of worry and concern over one's spending and financial security. The MAS shows good psychometric properties and relates to compulsive spending and credit card use among college students (Roberts & Jones, 2001).

Compulsive Buying Scale (CBS)

The CBS is a 7-item scale that was developed to identify compulsive buying (Faber & O'Guinn, 1992). Its items address behaviors specific to compulsive buying rather than normal consumer behavior (e.g., "Others would be horrified if they knew my spending habits" and "I bought things even though I couldn't afford them"). The scale shows good validity and reliability. It also correctly classified 87.5% of the subjects as compulsive buyers or normal consumers.

Results

Scale Construction and Factor Analyses of the EPFS

Construction of the scale began with generating 26 items that created from a literature review and examination of items of other self-rating executive function scales. The items were re-written, however, to pertain specifically to some aspect of personal finance. Items were rated on a Likert-type scale ranging from 1 (never) to 5 (always). Values for negative items were inverted so that higher scores represent better executive functioning. After the instrument was administered, corrected item-total correlations were obtained, which ranged from 0.33 to 0.67, $p < .001$. Exploratory factor analysis was done using principal components analysis with varimax rotation. The five factors accounted for 16.0%, 15.1%, 13.4%, and 11.9% of the variance, and cumulatively 56.4%.

Table 1: Factor analysis of the Executive Personal Finance Scale (n=225)

1	2	3	4	Item
+0.79	+0.05	-0.03	-0.02	When I see something I want, I have a hard time not buying it.
+0.78	-0.06	-0.14	-0.24	I buy things I don't really need
+0.70	-0.21	-0.17	+0.14	I tend to spend more when I know that I am already over my limit
+0.70	+0.00	-0.16	-0.12	I spend more money than I can afford to spend
+0.66	+0.06	-0.07	-0.14	When I go to a store, I end up buying things I didn't set out to buy
+0.62	-0.02	-0.12	-0.36	I tend to spend money as soon as I get it.
-0.07	+0.89	-0.01	+0.08	I work hard at making money.
-0.01	+0.87	+0.07	+0.12	I put a lot of effort into making money.
+0.05	+0.77	+0.02	+0.26	I work long hours to increase my wealth.
-0.04	+0.76	+0.22	-0.04	I'm enthusiastic when it comes to making money.
+0.03	+0.03	+0.74	+0.20	I am organized at balancing my checkbook.
+0.26	+0.08	-0.74	-0.13	I have trouble keeping my finances organized.
-0.07	+0.22	+0.70	+0.15	I keep my bills organized.
+0.26	-0.06	-0.69	+0.08	I make a lot of mistakes when keeping track of my money.
-0.11	+0.21	+0.59	+0.43	I have a system set up for managing my money.
-0.10	+0.06	+0.07	+0.81	I make investments for the long term.
+0.06	+0.07	-0.01	+0.71	I buy stocks and mutual funds for investment.
-0.30	+0.14	+0.20	+0.59	I put money into savings on a regular basis & leave it there.
-0.25	+0.21	+0.29	+0.57	I save money for the future.
<u>-0.24</u>	<u>+0.02</u>	<u>+0.34</u>	<u>+0.54</u>	<u>I set aside money for emergencies.</u>

High factor loadings are in bold face.

To shorten the scale, a maximum of 6 items with the highest loadings were chosen from each factor and the factor analysis was repeated using only those items. The final scale included a total of 20 items. Factor analysis again yielded a 4 factor structure. The four factors were named Impulse Control (IC), Motivational Drive (MD), Organization (ORG), and Planning (SP) based on the content of the items (Table 1). Cronbach alphas were good for the subscales: 0.80, 0.83, 0.86, 0.76, respectively, and 0.86 for the total score.

Demographic Influences on the EPFS

A linear regression was performed to determine the contributions of demographic variables to EPFS total score (Table 2). EPFS scores increased as a function of age and income, whereas sex and education had no apparent influence on the total score.

Table 2: Linear regression of demographic variables predicting total scores of the EPFS

	Beta	
Age	0.15	p<.05
Sex	0.09	
Education	0.04	
Income	0.28	p<.001

Construct Validity

Correlations were calculated between the EPFS and financial parameters (Table 3). Positive correlations were found between income and all scales of the EPFS. Significant correlations were found between EPFS scores and all financial behavior except for credit card debt. Controls for sex, age, education and income modified these correlations.

Table 3: Bivariate correlations between subscales of the EPFS and personal finances

	IC	MD	ORG	PL	Total
Income	+0.15*	+0.24†	+0.23†	+0.41†	+0.36†
Credit card debt current	-0.10	+0.11	-0.01	+0.08	+0.02
Credit card debt maximum	-.07	+0.11	+0.04	+0.09	+0.05
Debt/income ratio	-0.23**	-0.04	-0.11	-0.17*	-0.21**
Savings	+0.09	-0.02	+0.25†	+0.26†	+0.22†
CDs	+0.04	+0.01	+0.15*	+0.24†	+0.16*
Mutual funds	+0.12	+0.10	+0.16*	+0.36†	+0.27†
Stocks	+0.10	+0.30†	+0.20**	+0.43†	+0.36†
Retirement	+0.09	+0.22†	+0.20**	+0.45†	+0.34†
Real estate	+0.14*	+0.05	+0.19**	+0.25†	+0.23†

*p=.05, **p<.01, †p<.001

Discussion

This study demonstrated that executive aspects of managing personal finances factor together in a logical manner as would be expected from other studies of more general executive functions. These factors are impulse control (IC), organization (ORG), planning (PL), and motivational drive (MD). Linear regression indicated that executive financial management skill (total EPFS scores)

increases with age and income level, but does not vary according to sex or education. The validity for this scale is provided by correlations with individual income, and the tendency to use various forms of financial investment. Further validity for the scale is provided by significant correlations with other established scales measuring financial attitudes and behavior, the CBS and MAS.

Thus, the EPFS shows good psychometric properties and is short and easy to use. It relates logically to real-life management of personal finances and correlates well with other established measures of financial behavior. As such, the EPFS is a convenient measure to incorporate into future research using other methodologies such as objective neuropsychological tests, neuroimaging, and clinical populations.

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